CFG-GNN · SINGLE-BINARY INFERENCE

Behaviour and block localisation report

Function-level anti-debug, keylogger, and injection scores with basic-block localisation for one x86 PE.

Sample SHA-256	Architecture	Generated
25c5f0d873e8167464eab10187deb28da54656398d26c7dad9ff1918a25d3dd0	X86	2026-10-22 23:10 UTC

File: `data/raw/antidebug/0fe82dc1245fbb5001e1fae0396549fc50a083a82e1d63efa1f837813be691b6.exe`
Size: 71168 bytes · **Base:** `0x10000000` · **Functions:** 116 · **Blocks:** 1912 · **Edges:** 1674

ANTI_DEBUG
PRESENT
0.9989
1 / 116 functions
0 localized blocks

KEYLOGGER
absent
0.9227
0 / 116 functions
0 localized blocks

INJECTION
PRESENT
0.9996
21 / 116 functions
205 localized blocks

> **Interpretation.** "Absent" means no function crossed the target threshold; it is not a benign verdict. The model is trained on malware only and does not attribute a malware family. Blocks are model-ranked candidates, not manual ground truth.

Run metadata

Extraction	pre-extracted graph · 0.0s · 116 functions · 1912 blocks
API resolution	119 imports · 388 thunks · loaded=True
Stage 1 checkpoint	`/home/lali/cfg-gnn/results/train/stage1/best.pt`
Stage 2 checkpoint	`/home/lali/cfg-gnn/results/train/stage2/best.pt`
Stage 1 thresholds	[0.95, 0.94, 0.95]
Stage 2 thresholds	[0.94, 0.4999999999999994, 0.6399999999999999]

anti_debug PRESENT

Stage 1 threshold 0.9500 · Stage 2 threshold 0.9400 · flagged 1 / 116 functions

function VA	function RVA	function score	max block score	blocks above threshold
`0x10003e20`	`0x3e20`	0.9989	0.8085	0

Block localization — anti_debug

`0x3e20` score 0.9989

No block crossed the Stage 2 threshold. Best block score: **0.8085** (threshold 0.9400).

keylogger absent

Stage 1 threshold 0.9400 · Stage 2 threshold 0.5000 · flagged 0 / 116 functions

No function crossed the Stage 1 threshold for this target.

injection PRESENT

Stage 1 threshold 0.9500 · Stage 2 threshold 0.6400 · flagged 21 / 116 functions

function VA	function RVA	function score	max block score	blocks above threshold
0x10005880	0x5880	0.9996	0.9974	12
0x10006a40	0x6a40	0.9995	0.9974	13
0x10007330	0x7330	0.9995	0.9964	28
0x10006f40	0x6f40	0.9993	0.9964	5
0x10002ad0	0x2ad0	0.9976	0.9992	7
0x100054a0	0x54a0	0.9974	0.9902	6
0x10004950	0x4950	0.9967	0.9991	22
0x10006e90	0x6e90	0.9965	0.9956	6
0x100040e0	0x40e0	0.9964	0.9974	7
0x10006850	0x6850	0.9941	0.9682	4
0x10004600	0x4600	0.9930	0.9974	5
0x100053a0	0x53a0	0.9890	0.9762	11
0x100035e0	0x35e0	0.9847	0.9547	19
0x100082a0	0x82a0	0.9797	0.9681	8
0x10005060	0x5060	0.9796	0.9682	8
0x10005670	0x5670	0.9760	0.9774	8
0x10006de0	0x6de0	0.9725	0.9997	6
0x10006be0	0x6be0	0.9720	0.9532	9
0x10006950	0x6950	0.9639	0.9456	3
0x10008150	0x8150	0.9545	0.9558	7
0x10007ef0	0x7ef0	0.9520	0.9749	11

Block localization — injection

0x5880 score 0.9996

block RVA	block VA	size	score
0x5afe	0x10005afe	7	0.9974
0x5ad4	0x10005ad4	13	0.9887

block RVA	block VA	size	score
`0x5b05`	`0x10005b05`	13	0.9887
`0x5ae1`	`0x10005ae1`	21	0.9684
`0x5b12`	`0x10005b12`	21	0.9684
`0x5ca5`	`0x10005ca5`	21	0.9684
`0x5c39`	`0x10005c39`	19	0.9579
`0x59c6`	`0x100059c6`	21	0.9569
`0x5b3b`	`0x10005b3b`	24	0.9505
`0x5c04`	`0x10005c04`	40	0.9426
`0x58d6`	`0x100058d6`	32	0.9422
`0x5c30`	`0x10005c30`	6	0.6798

`0x6a40` score 0.9995

block RVA	block VA	size	score
`0x6aca`	`0x10006aca`	7	0.9974
`0x6b69`	`0x10006b69`	11	0.9956
`0x6bca`	`0x10006bca`	9	0.9679
`0x6b31`	`0x10006b31`	15	0.9472
`0x6b04`	`0x10006b04`	19	0.9428
`0x6a40`	`0x10006a40`	24	0.9405
`0x6b92`	`0x10006b92`	7	0.9356
`0x6b81`	`0x10006b81`	14	0.9271
`0x6b46`	`0x10006b46`	15	0.9206
`0x6b17`	`0x10006b17`	11	0.8688
`0x6a6a`	`0x10006a6a`	14	0.8623
`0x6a58`	`0x10006a58`	18	0.7639
`0x6abc`	`0x10006abc`	14	0.6531

`0x7330` score 0.9995

block RVA	block VA	size	score
`0x779d`	`0x1000779d`	9	0.9964
`0x76bc`	`0x100076bc`	14	0.9959
`0x778f`	`0x1000778f`	7	0.9938
`0x7729`	`0x10007729`	13	0.9829
`0x7359`	`0x10007359`	21	0.9754

block RVA	block VA	size	score
`0x78fc`	`0x100078fc`	26	0.9439
`0x7580`	`0x10007580`	19	0.9432
`0x799b`	`0x1000799b`	20	0.9422
`0x763c`	`0x1000763c`	11	0.9341
`0x79c4`	`0x100079c4`	18	0.9325
`0x77e9`	`0x100077e9`	69	0.9252
`0x76ca`	`0x100076ca`	58	0.9178
`0x747e`	`0x1000747e`	23	0.9158
`0x78b4`	`0x100078b4`	26	0.9149
`0x77c0`	`0x100077c0`	23	0.9148
`0x7549`	`0x10007549`	14	0.8962
`0x7601`	`0x10007601`	10	0.8955
`0x777f`	`0x1000777f`	16	0.8883
`0x74df`	`0x100074df`	38	0.8617
`0x7410`	`0x10007410`	40	0.8611
`0x7388`	`0x10007388`	14	0.8275
`0x743c`	`0x1000743c`	32	0.8218
`0x7509`	`0x10007509`	30	0.8214
`0x753a`	`0x1000753a`	9	0.7864
`0x746f`	`0x1000746f`	9	0.7864
`0x7862`	`0x10007862`	19	0.7075
`0x7534`	`0x10007534`	6	0.6798
`0x7469`	`0x10007469`	6	0.6798

`0x6f40` score 0.9993

block RVA	block VA	size	score
`0x711e`	`0x1000711e`	7	0.9964
`0x70ec`	`0x100070ec`	16	0.9480
`0x7091`	`0x10007091`	17	0.9409
`0x70ac`	`0x100070ac`	16	0.8249
`0x70c1`	`0x100070c1`	12	0.8046

0x2ad0 score 0.9976

block RVA	block VA	size	score
0x2c45	0x10002c45	8	0.9992
0x2b8e	0x10002b8e	20	0.9619
0x2ad0	0x10002ad0	32	0.9488
0x2b32	0x10002b32	12	0.9366
0x2b53	0x10002b53	48	0.9116
0x2b20	0x10002b20	18	0.8075
0x2c2f	0x10002c2f	12	0.7944

0x54a0 score 0.9974

block RVA	block VA	size	score
0x54c1	0x100054c1	10	0.9902
0x55b2	0x100055b2	13	0.9872
0x5657	0x10005657	7	0.9763
0x55bf	0x100055bf	32	0.9463
0x5520	0x10005520	88	0.9422
0x5592	0x10005592	21	0.8913

0x4950 score 0.9967

block RVA	block VA	size	score
0x4bcd	0x10004bcd	8	0.9991
0x4b24	0x10004b24	11	0.9813
0x4acf	0x10004acf	10	0.9802
0x4ad9	0x10004ad9	22	0.9582
0x4b63	0x10004b63	20	0.9489
0x4a1c	0x10004a1c	20	0.9489
0x49a6	0x100049a6	12	0.9476
0x4a52	0x10004a52	17	0.9442
0x4b99	0x10004b99	17	0.9442
0x4990	0x10004990	15	0.9409
0x4aef	0x10004aef	29	0.9346
0x4950	0x10004950	17	0.9224
0x49b2	0x100049b2	10	0.9197
0x4b10	0x10004b10	8	0.9182

block RVA	block VA	size	score
`0x496a`	`0x1000496a`	8	0.9139
`0x4972`	`0x10004972`	21	0.9129
`0x4a63`	`0x10004a63`	8	0.9098
`0x4a8f`	`0x10004a8f`	11	0.8894
`0x4bb3`	`0x10004bb3`	26	0.8242
`0x4a39`	`0x10004a39`	17	0.8124
`0x4b80`	`0x10004b80`	17	0.8124
`0x4b4a`	`0x10004b4a`	11	0.7607

`0x6e90` score 0.9965

block RVA	block VA	size	score
`0x6eea`	`0x10006eea`	7	0.9956
`0x6eb5`	`0x10006eb5`	13	0.9875
`0x6f26`	`0x10006f26`	7	0.9763
`0x6ea2`	`0x10006ea2`	19	0.9663
`0x6f11`	`0x10006f11`	14	0.9238
`0x6efa`	`0x10006efa`	15	0.8033

`0x40e0` score 0.9964

block RVA	block VA	size	score
`0x4218`	`0x10004218`	7	0.9974
`0x418c`	`0x1000418c`	13	0.9875
`0x413e`	`0x1000413e`	25	0.9554
`0x4377`	`0x10004377`	20	0.9460
`0x43a8`	`0x100043a8`	20	0.9240
`0x4210`	`0x10004210`	8	0.8282
`0x4394`	`0x10004394`	17	0.8172

`0x6850` score 0.9941

block RVA	block VA	size	score
`0x68a6`	`0x100068a6`	16	0.9682
`0x692f`	`0x1000692f`	14	0.9186
`0x6922`	`0x10006922`	13	0.8871
`0x68b6`	`0x100068b6`	15	0.8616

0x4600 score 0.9930

block RVA	block VA	size	score
0x4685	0x10004685	7	0.9974
0x4649	0x10004649	25	0.9554
0x46c3	0x100046c3	14	0.9212
0x4679	0x10004679	12	0.8318
0x4699	0x10004699	11	0.8252

0x53a0 score 0.9890

block RVA	block VA	size	score
0x5457	0x10005457	8	0.9762
0x5434	0x10005434	12	0.9754
0x5427	0x10005427	13	0.9637
0x53d2	0x100053d2	25	0.9555
0x53a0	0x100053a0	31	0.9467
0x5480	0x10005480	7	0.9430
0x5477	0x10005477	9	0.9387
0x5487	0x10005487	14	0.9295
0x53bf	0x100053bf	19	0.8709
0x53f9	0x100053f9	17	0.8486
0x5411	0x10005411	15	0.8365

0x35e0 score 0.9847

block RVA	block VA	size	score
0x38a7	0x100038a7	25	0.9547
0x37a5	0x100037a5	30	0.9534
0x3992	0x10003992	7	0.9411
0x3888	0x10003888	16	0.9368
0x3927	0x10003927	14	0.9284
0x38e5	0x100038e5	16	0.9282
0x35e0	0x100035e0	26	0.9272
0x372b	0x1000372b	19	0.9252
0x363c	0x1000363c	34	0.9165
0x3835	0x10003835	19	0.9155
0x35fa	0x100035fa	11	0.8797

block RVA	block VA	size	score
`0x38f5`	`0x100038f5`	19	0.8613
`0x396d`	`0x1000396d`	21	0.8455
`0x3908`	`0x10003908`	19	0.8447
`0x3982`	`0x10003982`	16	0.8432
`0x3958`	`0x10003958`	21	0.7986
`0x39c9`	`0x100039c9`	12	0.7428
`0x365e`	`0x1000365e`	11	0.6998
`0x3773`	`0x10003773`	6	0.6797

`0x82a0` score 0.9797

block RVA	block VA	size	score
`0x836f`	`0x1000836f`	10	0.9681
`0x8355`	`0x10008355`	11	0.9558
`0x83a6`	`0x100083a6`	14	0.9480
`0x8384`	`0x10008384`	14	0.9480
`0x82a0`	`0x100082a0`	38	0.9378
`0x82dd`	`0x100082dd`	17	0.9349
`0x8301`	`0x10008301`	66	0.9181
`0x8347`	`0x10008347`	10	0.7764

`0x5060` score 0.9796

block RVA	block VA	size	score
`0x5117`	`0x10005117`	16	0.9682
`0x5109`	`0x10005109`	14	0.9650
`0x516c`	`0x1000516c`	17	0.9432
`0x5060`	`0x10005060`	108	0.9408
`0x5147`	`0x10005147`	13	0.9327
`0x50fe`	`0x100050fe`	11	0.8855
`0x5134`	`0x10005134`	15	0.8261
`0x515c`	`0x1000515c`	16	0.8157

`0x5670` score 0.9760

block RVA	block VA	size	score
`0x5766`	`0x10005766`	13	0.9774
`0x5816`	`0x10005816`	8	0.9736

block RVA	block VA	size	score
`0x57cb`	`0x100057cb`	19	0.9579
`0x5670`	`0x10005670`	37	0.9461
`0x5797`	`0x10005797`	33	0.9434
`0x56aa`	`0x100056aa`	25	0.8481
`0x582e`	`0x1000582e`	36	0.7731
`0x57bc`	`0x100057bc`	6	0.6798

`0x6de0` score 0.9725

block RVA	block VA	size	score
`0x6e81`	`0x10006e81`	8	0.9997
`0x6e22`	`0x10006e22`	52	0.9535
`0x6de0`	`0x10006de0`	31	0.9503
`0x6e14`	`0x10006e14`	14	0.9475
`0x6e70`	`0x10006e70`	17	0.9432
`0x6dff`	`0x10006dff`	21	0.9049

`0x6be0` score 0.9720

block RVA	block VA	size	score
`0x6c09`	`0x10006c09`	28	0.9532
`0x6c70`	`0x10006c70`	14	0.9518
`0x6be0`	`0x10006be0`	27	0.9470
`0x6c47`	`0x10006c47`	13	0.9446
`0x6c64`	`0x10006c64`	7	0.9400
`0x6c32`	`0x10006c32`	12	0.9371
`0x6bfb`	`0x10006bfb`	14	0.8520
`0x6c54`	`0x10006c54`	16	0.7950
`0x6c2c`	`0x10006c2c`	6	0.6710

`0x6950` score 0.9639

block RVA	block VA	size	score
`0x6950`	`0x10006950`	33	0.9456
`0x69b5`	`0x100069b5`	14	0.9252
`0x6971`	`0x10006971`	15	0.8616

`0x8150` score 0.9545

block RVA	block VA	size	score
`0x8207`	`0x10008207`	11	0.9558
`0x825b`	`0x1000825b`	14	0.9480
`0x827d`	`0x1000827d`	14	0.9480
`0x8150`	`0x10008150`	35	0.9377
`0x818e`	`0x1000818e`	19	0.9311
`0x81b3`	`0x100081b3`	66	0.9180
`0x81f9`	`0x100081f9`	10	0.7764

`0x7ef0` score 0.9520

block RVA	block VA	size	score
`0x7f02`	`0x10007f02`	7	0.9749
`0x7ff8`	`0x10007ff8`	19	0.9642
`0x7f41`	`0x10007f41`	20	0.9504
`0x8070`	`0x10008070`	16	0.9480
`0x7f10`	`0x10007f10`	24	0.9385
`0x7f91`	`0x10007f91`	99	0.9039
`0x7f69`	`0x10007f69`	30	0.8907
`0x800f`	`0x1000800f`	27	0.8838
`0x8037`	`0x10008037`	31	0.8796
`0x7f55`	`0x10007f55`	20	0.8741
`0x805c`	`0x1000805c`	6	0.6698

Warnings

- PE machine field is 0; extraction/repair will determine the actual architecture

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Generated from schema `cfg-gnn.infer.v1`. The PDF lists Stage 1 selected functions; the JSON can additionally contain unselected function scores when `--all-functions` is used.